BRAIN-TO-VISION RECONSTRUCTION

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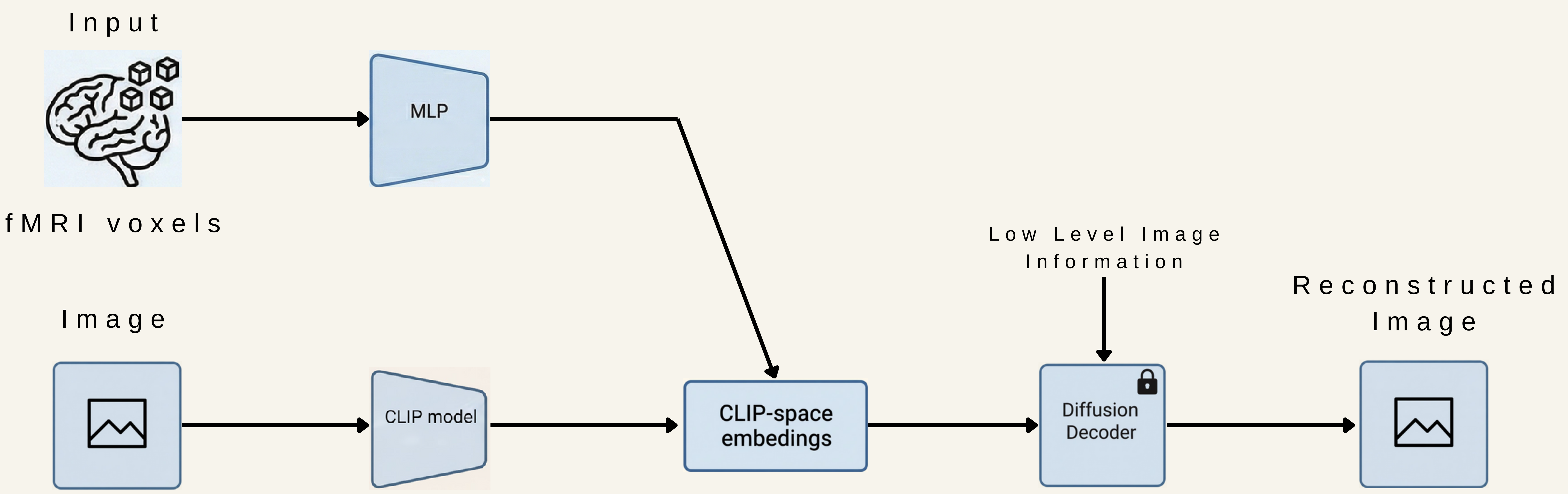
PROJECT IDEA / GENERAL OVERVIEW



MODEL ARCHITECTURE AND TRAINING

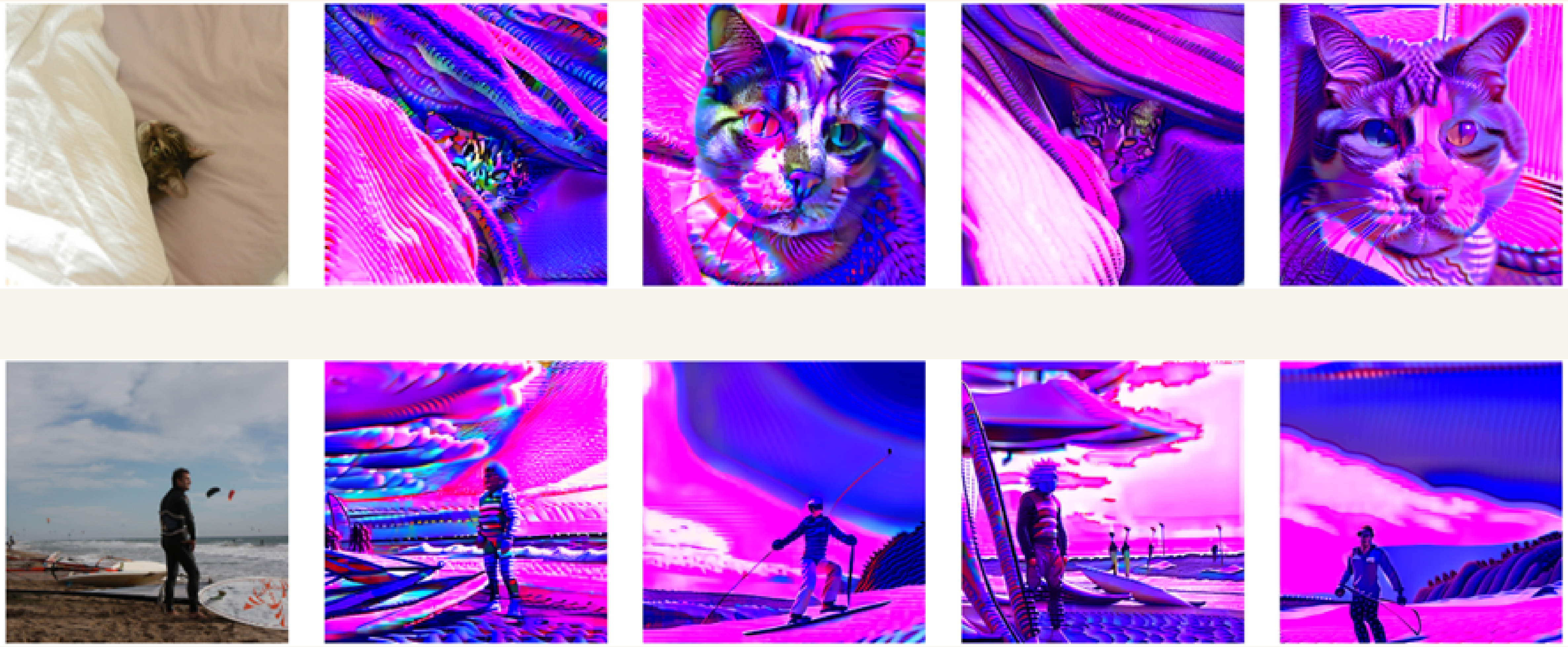
- **Deep funnel-shaped MLP encoder**, designed for dimensionality reduction of noisy fMRI signals
  - **Gradual reduction from 15,724 voxels to 1,280 dimensions**, compatible with the Kandinsky-CLIP space
  - **5 hierarchical linear layers** [12000→10000→8192→4096→2048] to gradually distill neural information
- **Combined Loss**: Triplet Loss + Contrastive Loss + Cosine Regression Loss
  - Goal: to **learn the structure of the semantic space**, not just minimize point-by-point error
  - The model **brings fMRI embeddings closer to the correct CLIP embeddings** and moves them away from semantically incorrect ones
- **20% dropout** and **Weight Decay** (**0.01**) to balance capacity and overfitting
- **Learning rate 5 x 10^{-4} with a 10-epoch warmup**, crucial for initial stability
- **High batch size (8192)** and training up to 50 epochs with convergence monitoring

DATASET ANALYSIS

- High-resolution fMRI from the **Natural Scenes Dataset (NSD)**, acquired with a 7 Tesla scanner
- Each **fMRI scan** is represented by **15,724 voxels** from the visual cortex
- Neural signals are paired with natural color images from the **COCO (Common Objects in Context)** dataset
- The analysis is focused on **Subject 1**, the most complete and commonly used subject in the MindEye literature
- We built an **aligned dataset**, in which each fMRI vector corresponds to a **1280-dimensional Kandinsky-CLIP embedding**
- The data were split at the image level into **training** (**8,000 unique images, 24,000 fMRI trials**) and **validation** (**2,000 unique images, 6,000 fMRI trials**) sets, ensuring that different repetitions of the same image were never shared across splits and preventing data leakage
- The **fMRI signals** are **pre-normalized** to ensure signal stability across sessions, while the **CLIP embeddings** have been **L2-normalized** to facilitate learning of semantic similarity in the CLIP space

RESULTS

Original Image | Reconstruction True Clip with anchor | Reconstruction Model Clip with anchor | Reconstruction True Clip | Reconstruction Model Clip



Kandinsky Diffusor has a blue shift due to color bias in the training set, however, end-to-end training with reconstruction loss (as performed in the original paper) significantly improves the results

CONCLUSIONS

- **What we have learned**
  - fMRI signals are highly noisy, normalizing CLIP embeddings is crucial for convergence
  - The deep layers are mainly used for dimensional alignment, not for extracting new semantic features
  - The overall semantics are correct, but the spatial localization of objects may be inaccurate
- **Limits and future directions**
  - Reconstruction without low-level signals loses spatial structure
  - Low-level explicit conditioning is needed (e.g., outlines or spatial maps)
  - Next Step: Implement a “Low-level Adapter” (like ControlNet) trained on the voxels of the primary visual cortex (V1) to fix the spatial structure